

CRM Application for Jewel Management - Developer

Project Title: CRM Application for Jewel Management - Developer

Date: November 02, 2025

Team ID: NM2025TMID07062

Maximum Marks: 4 Marks

Phase 5: Project Design Phase II - Solution Requirements

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Customer Registration & Profile Management	Customer registration through intuitive form; Social login integration (Google/Facebook); Comprehensive profile with contact details, personal preferences, jewelry sizing information, complete purchase history
FR-2	Customer Portal	Secure multi-factor login capabilities; Comprehensive purchase history viewing; Real-time custom order tracking; Preference and profile updates; Convenient appointment scheduling; Receipt and certificate downloads
FR-3	Inventory Item Management	Add jewelry items with unique SKU, detailed descriptions, complete specifications; Track metal type and purity standards, gemstone details (4Cs: Carat, Cut, Clarity, Color), precise weight calculations, certification numbers; Upload multiple high-resolution photos; Assign acquisition cost and retail pricing; Set comprehensive status indicators (available, reserved, sold, consignment, in-repair)
FR-4	Inventory Tracking	Real-time stock level visibility across multiple store locations; Barcode and RFID scanning for efficient check-in/check-out operations; Location transfer tracking with audit trail; Automated intelligent reorder alerts; Advanced loss prevention tracking and reporting
FR-5	Inventory Valuation	Automated valuation based on current market prices; Historical valuation tracking for trend analysis; Consignment inventory separately tracked and managed; Insurance valuation reports generation
FR-6	Custom Order Management	Create custom orders with comprehensive design specifications; Customer approval workflows for design verification; Production milestone tracking and updates; Artisan assignment and workload management; Quality inspection checkpoints enforcement; Delivery scheduling coordination
FR-7	Custom Order Communication	Automated milestone notifications at each stage; Customer portal for real-time progress viewing; Design modification request handling; Photo updates during production phases; Transparent communication channels
FR-8	Sales Pipeline & Opportunities	Systematic lead capture from walk-ins and online inquiries; Opportunity tracking organized by defined pipeline stages; Intelligent lead scoring based on engagement metrics and budget indicators; Comprehensive win/loss analysis reporting

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Quote & Pricing	Dynamic quote generation with configurable pricing rules; Integrated flexible financing options; Trade-in valuation processing; Comprehensive discount management; Quote expiration and renewal tracking
FR-10	Sales Order Processing	Seamless conversion of quotes to sales orders; Integrated payment processing capabilities; Detailed receipt generation with itemization; Automated tax calculation; Warranty and certificate generation
FR-11	Marketing Automation	Automated birthday and anniversary reminder campaigns; Timely new collection announcements; Segment-based targeted promotional campaigns; Referral program tracking and management; VIP customer recognition programs
FR-12	Customer Segmentation	Sophisticated segmentation by purchase history, preferences, budget tier; Behavioral segmentation based on purchase patterns; Automated inactive customer identification; High-value customer identification and prioritization
FR-13	Data Protection & Business Rules	Prevent unauthorized deletion of customers with active orders; Prevent deletion of inventory in pending transactions; Prevent deletion of historical sales for compliance requirements; Comprehensive audit trail for all data modifications
FR-14	Security & Access Control	Role-based permissions (sales staff, managers, owners, administrators); Multi-factor authentication for sensitive operations; Robust data encryption for customer and financial information; Advanced session management and automatic timeout protocols
FR-15	Dashboards & Analytics	Sales performance visualization by time period, category, and staff; Inventory valuation and turnover analytics; Customer lifetime value calculations; Marketing campaign effectiveness metrics; Detailed profit margin analysis
FR-16	Mobile Access	Mobile-optimized responsive interface; Comprehensive customer profile access on mobile; Real-time inventory availability checking; Mobile transaction processing; Customer photo and signature capture capabilities
FR-17	Reporting	Customizable report generation; Scheduled automated report delivery; Export functionality to Excel and PDF formats; Tax compliance reporting capabilities; Insurance and customs documentation generation

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive interface specifically designed for non-technical jewelry retail staff. Mobile application must facilitate easy use during customer consultations. Comprehensive training materials and context-sensitive in-app help available. Maximum 2 hours training required for competent basic operations. Accessibility compliance for users with varying abilities.
NFR-2	Security	All sensitive data (customer information, inventory valuations, financial transactions) encrypted at rest and in transit using industry-standard protocols. Multi-factor authentication mandated for administrative functions. Role-based access control strictly enforced with regular permission audits. Full PCI-DSS compliance for payment data handling. Annual comprehensive security audits and penetration testing.

NFR No.	Non-Functional Requirement	Description
NFR-3	Reliability	Guaranteed 99.9% system uptime throughout operational year. Automated comprehensive backups executed every 4 hours. Sophisticated data validation rules preventing incorrect entries. Business rule enforcement ensuring complete data integrity. Zero tolerance policy for data loss in high-value transactions. Graceful error handling with clear user-friendly error messages.
NFR-4	Performance	Page load times consistently maintained below 2 seconds for all standard operations. Inventory search results displaying in under 1 second for large datasets. Mobile application responsive performance across 3G network connections. Support for 50+ concurrent users without degradation. Database queries fully optimized for large inventory management (10,000+ items efficiently).
NFR-5	Availability	Customer portal accessible 24/7/365 without interruption. Maintenance windows strategically scheduled during off-peak hours (midnight-3am local time). Disaster recovery capabilities with Recovery Time Objective (RTO) of less than 2 hours. Automated failover mechanisms for all critical services. Load balancing across multiple servers for optimal resource utilization.
NFR-6	Scalability	Support deployment from single-location boutiques to 50+ store chains without architectural changes. Handle inventory growth from initial 100 items to 100,000+ comprehensive items. Scale seamlessly to manage millions of customer records. Accommodate seasonal traffic spikes during holiday periods. Cloud infrastructure auto-scaling for dynamic resource allocation.
NFR-7	Maintainability	Modular system design enabling easy feature additions and modifications. Comprehensive technical documentation for system administrators and developers. Version control integration for all customizations and code changes. Automated testing framework for core business workflows. Streamlined upgrade path supporting Salesforce platform updates and patches.
NFR-8	Compliance	Full PCI-DSS compliance certification for payment processing operations. Complete GDPR compliance for customer data collection, storage, and processing. Automated sales tax calculation accuracy across all applicable jurisdictions. Comprehensive audit trails for regulatory compliance and reporting requirements. Support for insurance and customs documentation generation.
NFR-9	Compatibility	Full support for iOS and Android mobile platforms with optimized user experiences. Browser compatibility across Chrome, Firefox, Safari, and Edge. Seamless integration with common POS (Point of Sale) systems. Support for barcode scanners and RFID readers from major manufacturers. REST API integration with appraisal services and gemstone certification databases.
NFR-10	Data Quality	Validation rules enforcing mandatory field completion. Format validation for SKUs, certification numbers, and identification codes. Automated duplicate detection for customer and inventory records. Data cleansing tools for bulk imports and migrations. Historical data retention for 7+ years supporting regulatory and business requirements.
NFR-11	Disaster Recovery	Daily automated backups replicated to geographically distributed cloud locations. Backup retention policy maintaining data for minimum 1 year. Recovery Time Objective (RTO) guarantee of 2 hours maximum. Recovery Point Objective (RPO) of 4 hours maximum data loss acceptable window. Quarterly comprehensive disaster recovery testing and validation drills.
NFR-12	Internationalization	Multi-currency support with automated real-time conversion rates. Multi-language interface supporting English, Spanish, Chinese, and Hindi. Regional tax and compliance rules specific to each market. Support for varying jewelry metal purity standards (karat vs. fineness notations). Cultural date and time format adaptations.

System Architecture & Technical Specifications

Technology Stack

Cloud Platform: Salesforce Sales Cloud + Experience Cloud for comprehensive CRM and customer engagement capabilities

Frontend: Lightning Web Components (LWC) for responsive web interface; Native Salesforce Mobile App with offline capabilities

Backend: Salesforce Apex classes, REST APIs, and batch processing for complex operations

Database: Salesforce multi-tenant database with custom objects optimized for jewelry retail operations

Automation: Salesforce Flow Builder, Process Builder, Approval Processes, and scheduled actions

Integration: RESTful APIs for POS systems, payment gateways, appraisal services, and certification databases

Security: Salesforce Shield for advanced encryption, event monitoring, and field-level audit trails

Mobile: Salesforce Mobile SDK with offline synchronization capabilities for seamless mobile operations

Data Model & Custom Objects

Customer: Standard Contact object extended with jewelry-specific fields including preferences, precise sizing information, VIP status, communication preferences, jewelry allergies, metal sensitivities

Jewelry_Item: Comprehensive tracking including SKU, category classification, metal type and purity standards, primary and secondary stone specifications, precise weight, certification numbers, multiple high-resolution photos, acquisition cost, current retail pricing, detailed status tracking, store location, supplier information

Custom_Order: Customer reference, detailed design specifications and requirements, design approval status, production milestone tracking, assigned artisan and team, estimated completion date, actual completion status, cost tracking, delivery information

Sales_Opportunity: Standard Opportunity object enhanced with jewelry-specific fields including product category, budget confirmation, trade-in valuation, financing requirements, special requests

Appraisal: Item reference, qualified appraiser information, appraisal date, valuation amount, appraisal purpose (insurance/resale), certification documentation

Marketing_Campaign: Standard Campaign object with jewelry-specific segmentation criteria, personalization fields, response tracking for fine-tuned targeting

Key Business Rules

BR-ID	Business Rule Description
BR-1	Prevent deletion of customers with active orders or orders within the past 2 years to maintain historical data integrity
BR-2	Prevent deletion of inventory items with pending transactions or items sold within the past 7 years for compliance and audit purposes
BR-3	Require manager-level approval for all discounts exceeding 15% to maintain pricing controls
BR-4	Automated alerts triggered when inventory item has not been scanned for 90+ days (advanced loss prevention mechanism)
BR-5	Automated reorder alerts triggered when category stock levels fall below established minimum thresholds
BR-6	Automated customer notification sent when custom order reaches each defined production milestone
BR-7	Require certification number validation for all gemstones exceeding 0.5 carats
BR-8	Comprehensive audit log maintained for all changes to inventory valuations and customer financial data

Acceptance Criteria

Requirement Category	Specific Acceptance Criteria
FR-1 to FR-2	Customer registration and portal features fully functional with 100+ test customers; Portal accessible across web and mobile platforms; Login authentication working reliably
FR-3 to FR-5	500+ jewelry items successfully entered with complete specifications; Barcode scanning operational across all store locations; Real-time inventory valuation updating correctly
FR-6 to FR-7	50+ custom orders tracked through entire production workflow; Automated notifications sent at each milestone; Customer portal accurately displaying progress information
FR-8 to FR-10	Sales pipeline populated with 100+ qualified opportunities; Quote generation functionality operational and tested; Payment processing integrated and secure
FR-11 to FR-12	Automated campaigns successfully sent to segmented customer lists; Email open rates exceeding 25%; Click-through rates exceeding 5%; Campaign analytics tracking accurately
FR-13 to FR-14	Business rules preventing 100% of unauthorized deletion attempts; Role-based permissions strictly enforced; Zero security breaches or unauthorized access incidents
FR-15 to FR-17	Dashboards loading within 2 seconds; Mobile application fully functional on iOS and Android; Reports exportable to all required formats; No data integrity issues
NFR-1 to NFR-4	90% of staff trained to competency within 2 hours; 99.9% uptime achieved consistently; All pages loading within 2 seconds; Zero data loss incidents
NFR-5 to NFR-8	System scales to 10 locations without performance degradation; PCI-DSS compliance certified by third party; Complete audit trails maintained for all transactions
NFR-9 to NFR-12	Mobile app functional on iOS and Android; POS system integration successful; Multi-currency support operational; Backup and recovery systems tested and validated

Constraints & Dependencies

Project Constraints

- **Budget Limitation:** Mid-market jewelry retailer budget parameters require cost-effective implementation
- **Development Timeline:** 46-day aggressive timeline for complete development and deployment
- **Team Composition:** 3 developers with varying levels of Salesforce platform expertise requiring knowledge sharing
- **Regulatory Compliance:** Mandatory compliance with PCI-DSS and GDPR regulations throughout implementation
- **Infrastructure:** Existing IT infrastructure and network bandwidth constraints

Project Dependencies

- **Salesforce Platform:** Adherence to Salesforce API governor limits and service limits
- **Third-Party Integrations:** Availability and compatibility of POS systems, payment gateways, and appraisal service providers
- **Hardware Integration:** Availability and compatibility of barcode scanners and RFID readers from multiple manufacturers
- **User Adoption:** Critical success factor of staff training effectiveness and organizational change management
- **Data Migration:** Quality and completeness of legacy data from existing retail systems requiring cleansing

Conclusion

This comprehensive phase establishes detailed functional and non-functional requirements specifications for the jewelry management CRM application. The 17 functional requirements and 12 non-functional requirements address the unique operational challenges inherent in jewelry retail operations including sophisticated inventory management, custom order workflows, personalized customer relationships, and stringent security and compliance needs. With clear acceptance criteria, detailed technical specifications, and identified constraints and dependencies, the development team has a precise roadmap for delivering a solution that measurably improves operational efficiency, significantly enhances customer satisfaction, and drives sustainable business growth for jewelry retailers of all sizes.